

Affordance and other topics

(physical design and more...)

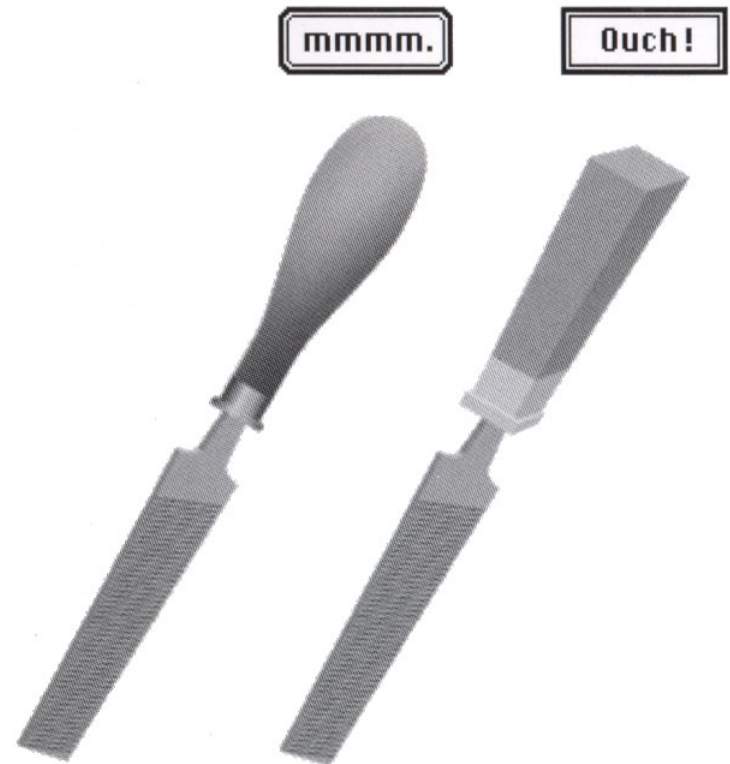
Affordance (*Invito all'uso*)

Affordance: (invitation to use) defines the physical quality of an object that suggests to a human being the appropriate actions to manipulate it.



Affordances

Affordances "suggest" how an object may be
interacted with



Affordances of buttons

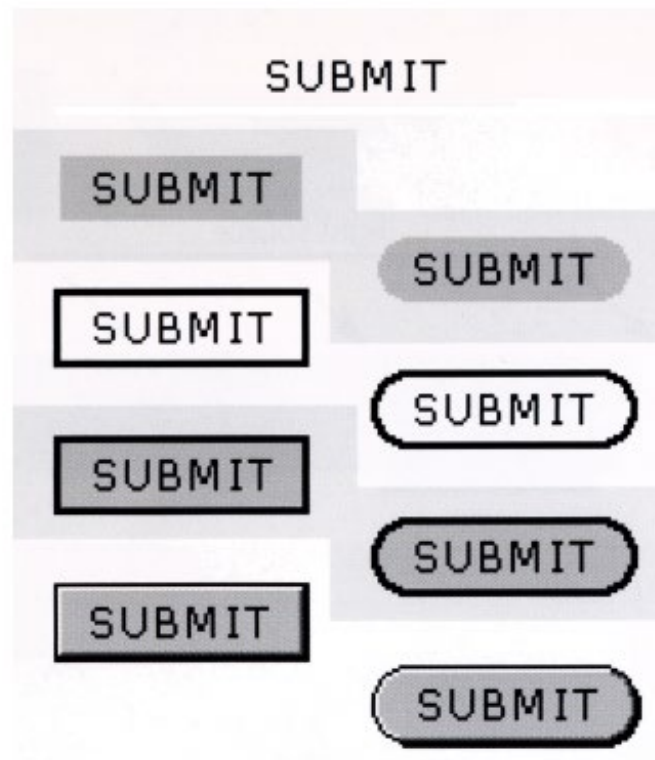
no affordances

rectangle

border

outline

3d



not clickable

less clickable

more clickable



Affordances



knurls
(*zigrinature*)



Button - Push



Switch - Flip



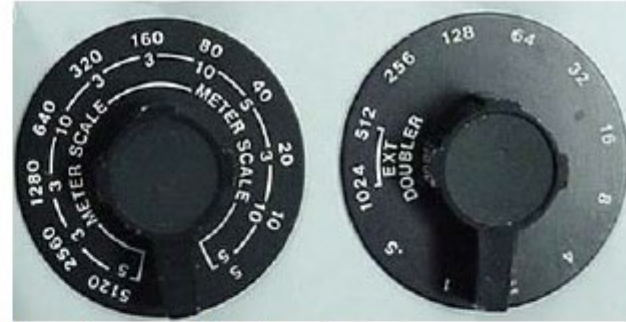
Knob - Rotate



Affordances



chairs are for sitting
table for placing things on



knobs are for turning



slots are for inserting
handles are for turning



buttons are for pressing



switch for toggling



dials for turning

sliders for sliding

Problems with affordance

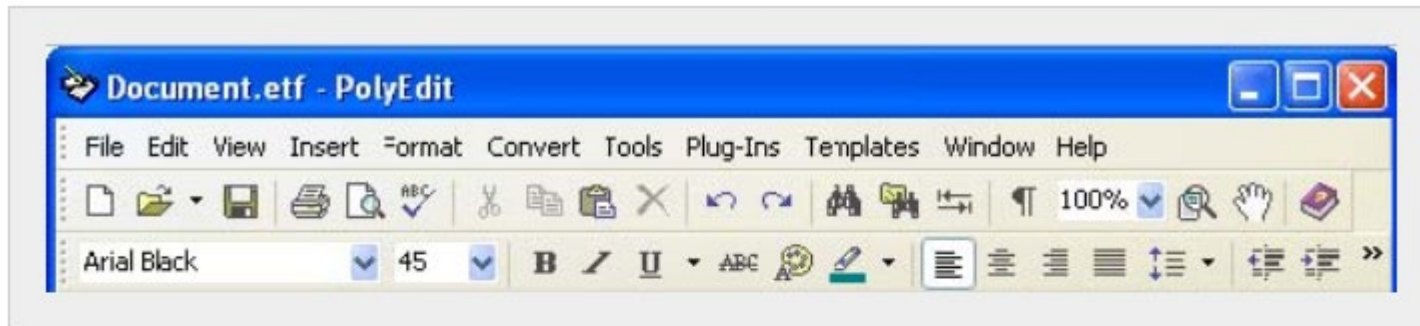
is this equalizer control a toggle or button?

button for pressing, but action unknown

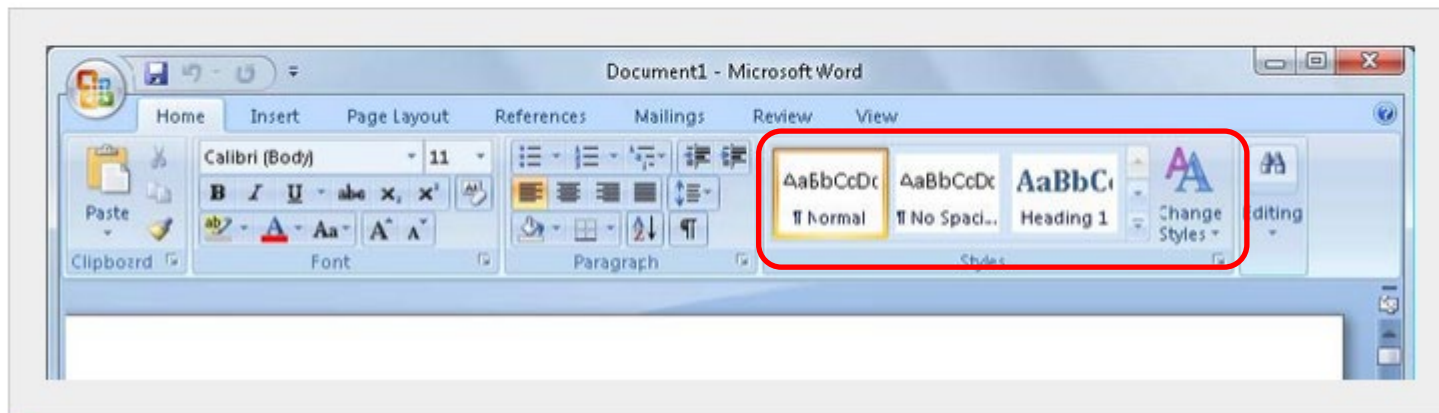
are these buttons?



Problems with affordance



❌ Figure: Bad Example - Old MS Word - Because of the UI, people never knew they could use styles e.g. normal, H1, H2



✅ Figure: Good Example - New MS Word - Because of the new ribbon UI, people intuitively know how to use styles

Problems with affordance



The fact that the staff of the store had to modify the machine in this way gives us a clue that there may be a problem with the design. It suggests that some customers were holding their cards in the wrong place..

Physical design



in the kitchen:
buttons are smooth, and easy to clean

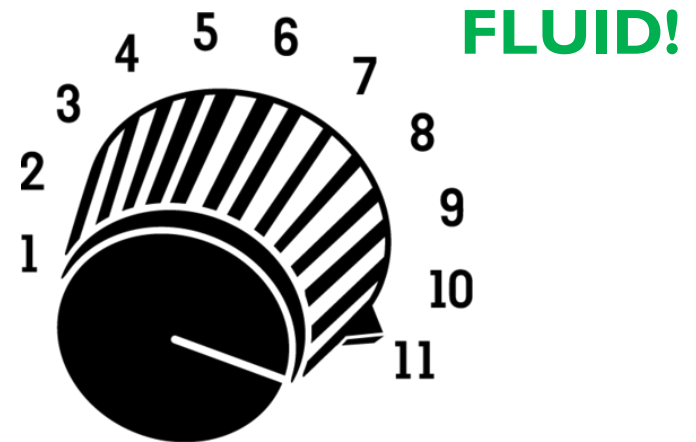
Fluidity

Fluidity: the extent to which the **physical structure and manipulation** of the device **naturally** relate to the **logical functions** it supports

Three checkpoints:

I. Do external physical aspects reflect logical effect?

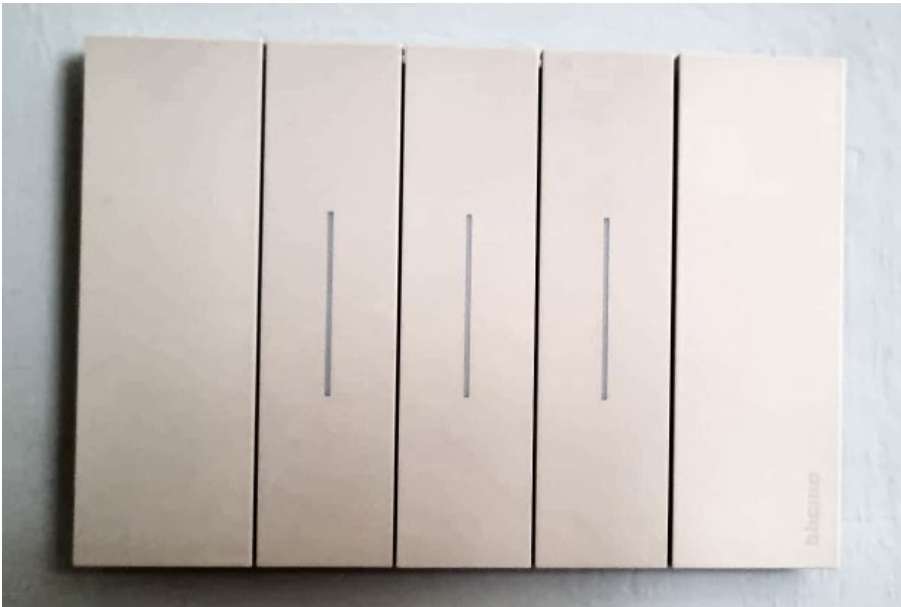
(what does it mean to turn the knob?)



Fluidity

Three checkpoints:

2. Is the logical state revealed in physical state?
(e.g. on/off buttons)



FLUID!



NOT FLUID (without led)

FLUID (with led)

Fluidity

Three checkpoints:

3. Do inverse actions correspond to inverse effects?
(e.g. arrow buttons, twist controls)

FLUID!



Physical design: constraints

Ergonomic (e.g. minimum button size)

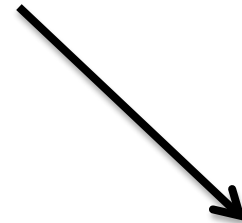
Physical (e.g. high-voltage switches are big)

Legal and safety (e.g. cooker controls..)

Context and environment
(e.g. easy to clean controls)

Aesthetic (must look good)

Economic (...and not cost too much!)



Physical design: trade offs

Constraints are **contradictory** ... need trade-offs

Contradiction **between categories**:

e.g. ergonomics vs. physical – watch

ergonomics – controls need
to be bigger

physical – no room!

solution – multifunction controls



Contradiction **within categories**:

e.g. safety – cooker controls

front panel – safer for adult

rear panel – safer for child



Error and repair

Can't always avoid errors ...

... but we can put them right

Make it easy to detect errors!

... then the user can repair them

Hello, this is the **Go Faster** booking system. What would you like?

(user) I want to fly from New York to London

You want a ticket from New York to Boston

(user) no

Sorry, please confirm one at a time;
do you want to fly from New York

(user) yes

